

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which of the following expressions is equivalent to the sum of $(r^3 + 5r^2 + 7)$ and $(r^2 + 8r + 12)$?

Answer

- A. $r^5 + 13r^3 + 19$
- B. $2r^3 + 13r^2 + 19$
- C. $r^3 + 5r^2 + 7r + 12$
- D. $r^3 + 6r^2 + 8r + 19$

Correct Answer: D

Rationale

Choice D is correct. Grouping like terms, the given expressions can be rewritten as $r^3 + (5r^2 + r^2) + 8r + (7 + 12)$. This can be rewritten as $r^3 + 6r^2 + 8r + 19$.

Choice A is incorrect and may result from adding the two sets of unlike terms, r^3 and r^2 as well as $5r^2$ and $8r$, and then adding the respective exponents. Choice B is incorrect and may result from adding the unlike terms r^3 and r^2 as if they were r^3 and r^3 and adding the unlike terms $5r^2$ and $8r$ as if they were $5r^2$ and $8r^2$. Choice C is incorrect and may result from errors when combining like terms.